

Salesforce CRM Project Documentation

WhatNext Vision Motors: Shaping the Future of Mobility with Innovation and Excellence

Project Overview

WhatNext Vision Motors is a customized Salesforce CRM solution developed for managing automotive-related business operations.

The application provides a common workspace for handling information associated with:

- Vehicles
- Customers
- Dealers
- Vehicle Orders
- Test Drives
- Service Requests

The ordering process is an important part of the application. Vehicle availability is considered before an order is confirmed, and the system can automatically associate an appropriate dealer with an order based on the configured business logic.

The project also uses automation for activities such as test-drive reminders and processing pending orders. The Lightning interface provides users with organized navigation and record pages, while reports and dashboards support business analysis.

Project Objectives

The project has been developed with the following objectives:

1. Centralize Automotive Information

Create a common Salesforce environment for maintaining vehicle, customer, dealer, order, test-drive, and service information.

2. Improve Order Processing

Provide a structured method for creating and managing vehicle orders while considering vehicle availability.

3. Reduce Manual Activities

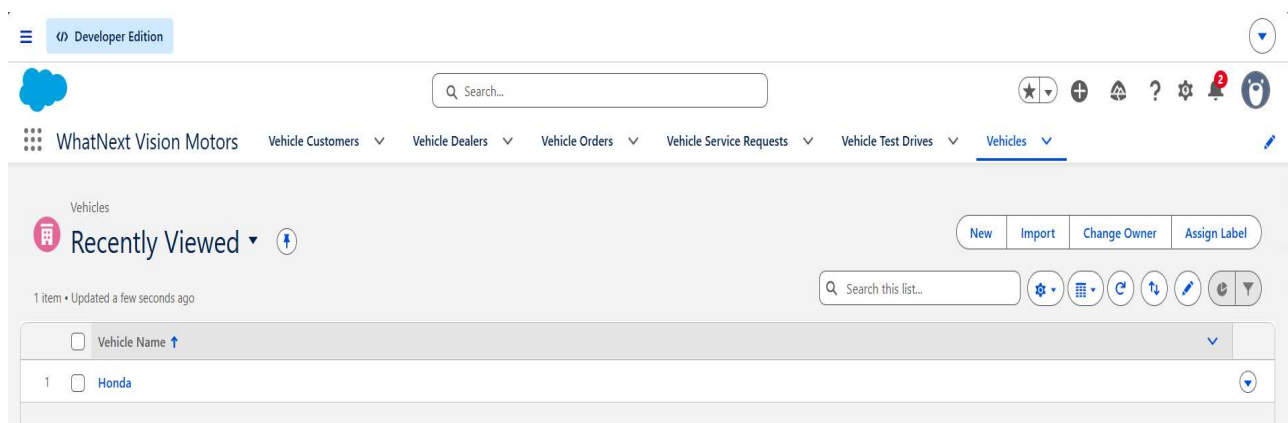
Automate repetitive operations using Salesforce Flow and Apex where appropriate.

4. Maintain Data Accuracy

Use validation rules and programmatic checks to reduce incorrect or incomplete transactions.

5. Simplify User Interaction

Create a Lightning application with organized navigation, page layouts, and relevant record information.



Phase 1: Requirement Analysis and Planning

The first phase concentrated on converting the business scenario into Salesforce requirements. The guidelines identify requirement analysis, project scope, data modeling, and security modeling as key activities in this phase.

1.1 Business Requirements

The requirements were identified by considering the activities performed by an automotive organization.

The system needs to maintain information about vehicles and customers while also supporting activities performed before and after a vehicle purchase.

The major business requirements identified were:

1. Store vehicle information.
2. Maintain customer records.
3. Maintain dealer information.
4. Record vehicle orders.
5. Associate orders with customers, vehicles, and dealers.

1.2 Project Scope

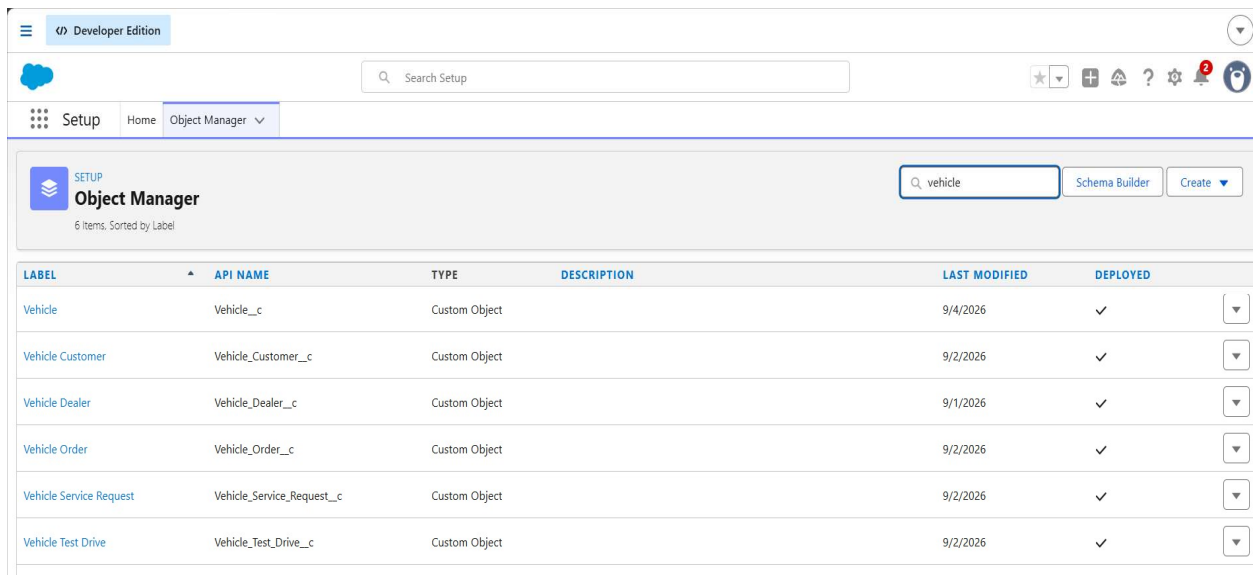
- **Vehicle Management:** Maintains essential vehicle information, including model details, pricing, stock availability, and related vehicle data.
- **Dealer Management:** Stores and manages dealership information to support vehicle sales and dealer-related activities.
- **Customer Management:** Maintains customer details and provides the required information for sales, test drives, and service interactions.
- **Order and Test Drive Management:** Records vehicle purchase orders and manages scheduled test-drive activities by connecting customers with their selected vehicles.
- **Service Request Management:** Records customer service requirements and connects service-related information with the appropriate customer and vehicle records.

1.3 Data Model

The data model represents the major entities involved in the automotive CRM process.

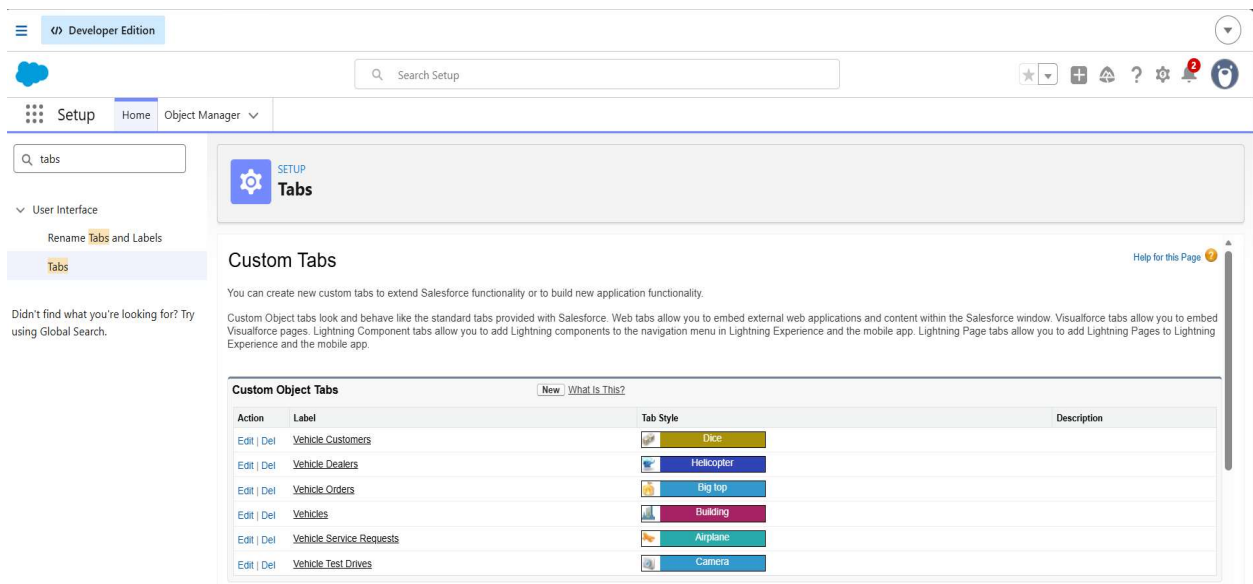
A vehicle can be associated with order and test-drive records. Customer information can be connected to orders and test drives. Dealer information can be associated with vehicle transactions.

The relationships reduce the need to duplicate information and make related records easier to access.



The screenshot shows the Salesforce Object Manager interface. At the top, there's a search bar with "vehicle" entered. Below the search bar, there's a table with 6 items, sorted by label. The table has columns for LABEL, API NAME, TYPE, DESCRIPTION, LAST MODIFIED, and DEPLOYED. The objects listed are Vehicle, Vehicle Customer, Vehicle Dealer, Vehicle Order, Vehicle Service Request, and Vehicle Test Drive, all of which are Custom Objects and have been deployed.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Vehicle	Vehicle__c	Custom Object		9/4/2026	✓
Vehicle Customer	Vehicle_Customer__c	Custom Object		9/2/2026	✓
Vehicle Dealer	Vehicle_Dealer__c	Custom Object		9/1/2026	✓
Vehicle Order	Vehicle_Order__c	Custom Object		9/2/2026	✓
Vehicle Service Request	Vehicle_Service_Request__c	Custom Object		9/2/2026	✓
Vehicle Test Drive	Vehicle_Test_Drive__c	Custom Object		9/2/2026	✓



The screenshot shows the Salesforce Tabs configuration page. On the left, there's a sidebar with a search bar and a list of tabs. The main content area is titled "Custom Tabs" and contains a table with columns for Action, Label, Tab Style, and Description. The table lists several tabs: Vehicle Customers (Dice), Vehicle Dealers (Helicopter), Vehicle Orders (Big Top), Vehicles (Building), Vehicle Service Requests (Airplane), and Vehicle Test Drives (Camera).

Action	Label	Tab Style	Description
Edit Del	Vehicle Customers	Dice	
Edit Del	Vehicle Dealers	Helicopter	
Edit Del	Vehicle Orders	Big Top	
Edit Del	Vehicles	Building	
Edit Del	Vehicle Service Requests	Airplane	
Edit Del	Vehicle Test Drives	Camera	

1.4 Security Model

Security was considered as part of the application design.

The Salesforce security model uses multiple levels of access control, including:

- Profiles
- Roles
- Role hierarchy
- Permission sets
- Object permissions

Phase 2: Salesforce Development - Backend & Configurations

The second phase involved implementing the required Salesforce data structures and backend functionality. The project guidelines specifically require documentation of objects, fields, validation, automation, Apex and related backend configurations where applicable.

2.1 Setup environment & DevOps workflow

The application was developed using Salesforce Lightning Experience.

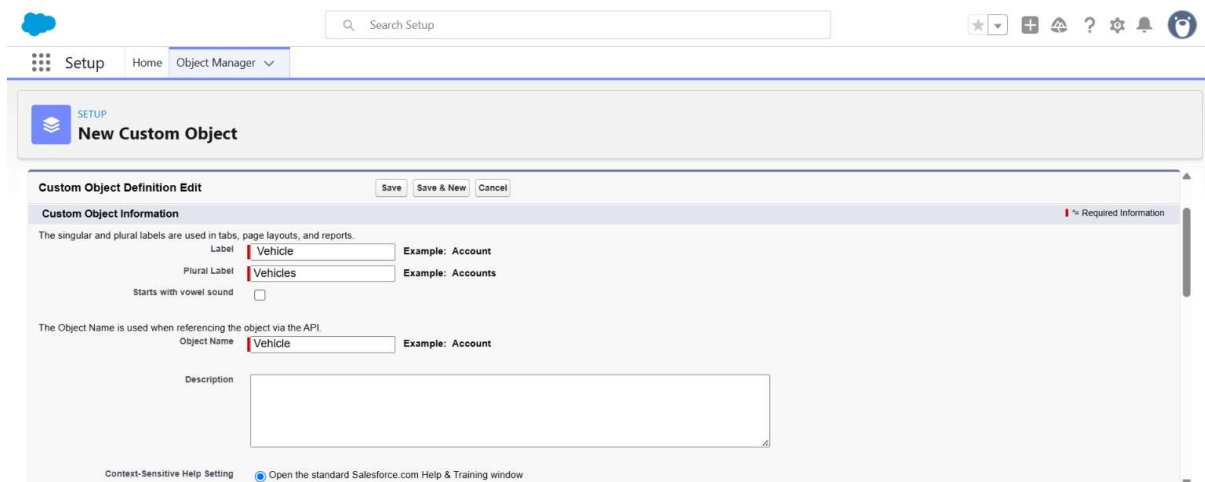
Salesforce provided the environment for creating custom objects, configuring fields, defining relationships, implementing automation, developing Apex components, creating reports, and managing security.

2.2 Customization of Objects, Fields, Validation Rules, Automation

2.2.1 Custom Objects

The project uses six custom objects to represent the major business entities of the WhatNext Vision Motors automotive CRM system:

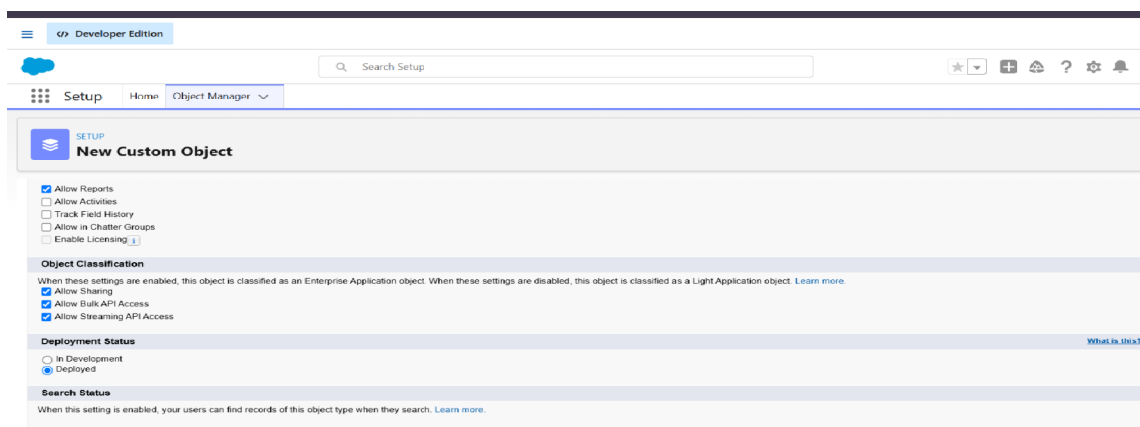
- **Vehicle:** Stores vehicle details such as model, price, stock quantity, availability, and dealer information.
- **Dealer:** Maintains dealership details, including dealer name, location, city, and contact information.
- **Customer:** Stores customer information required for vehicle purchases, test drives, and service activities.
- **Vehicle Order:** Records customer vehicle purchases and maintains details related to the selected vehicle, customer, dealer, and order status.
- **Vehicle Test Drive:** Manages test-drive bookings by maintaining customer, vehicle, appointment date, and status details.
- **Vehicle Service Request:** Records customer service requirements and maintains the related vehicle, customer, and service-status information.



The screenshot shows the 'New Custom Object' page in Salesforce Setup. The page title is 'New Custom Object'. Below the title, there are tabs for 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' tab is selected. The main content area is titled 'Custom Object Definition Edit' and includes a 'Save' button, a 'Save & New' button, and a 'Cancel' button. The 'Custom Object Information' section contains the following fields:

- Label:** Vehicle (Example: Account)
- Plural Label:** Vehicles (Example: Accounts)
- Starts with vowel sound:** ☐
- Object Name:** Vehicle (Example: Account)
- Description:** (Empty text area)

At the bottom of the page, there is a 'Context-Sensitive Help Setting' section with a link to 'Open the standard Salesforce.com Help & Training window'.



The screenshot shows the 'New Custom Object' page in Salesforce Setup, displaying additional settings. The page title is 'New Custom Object'. Below the title, there are tabs for 'Setup', 'Home', and 'Object Manager'. The 'Object Manager' tab is selected. The main content area is titled 'Custom Object Definition Edit' and includes a 'Save' button, a 'Save & New' button, and a 'Cancel' button. The 'Custom Object Information' section contains the following fields:

- Allow Reports:** ☒
- Allow Activities:** ☐
- Track Field History:** ☐
- Allow in Chatter Groups:** ☐
- Enable Licensing:** ☐

The 'Object Classification' section contains the following fields:

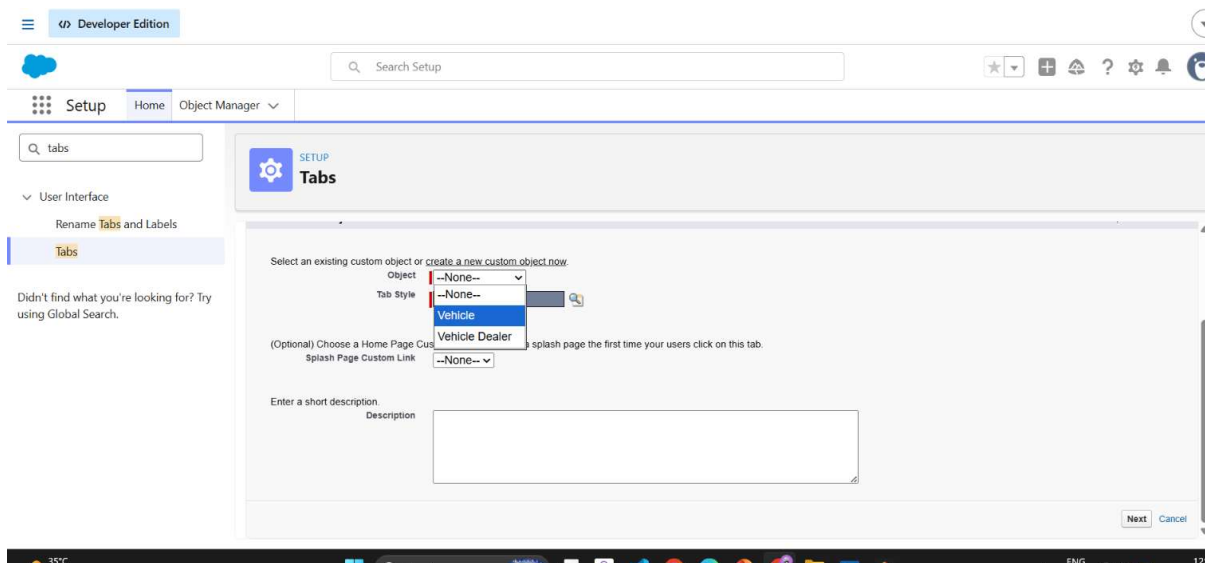
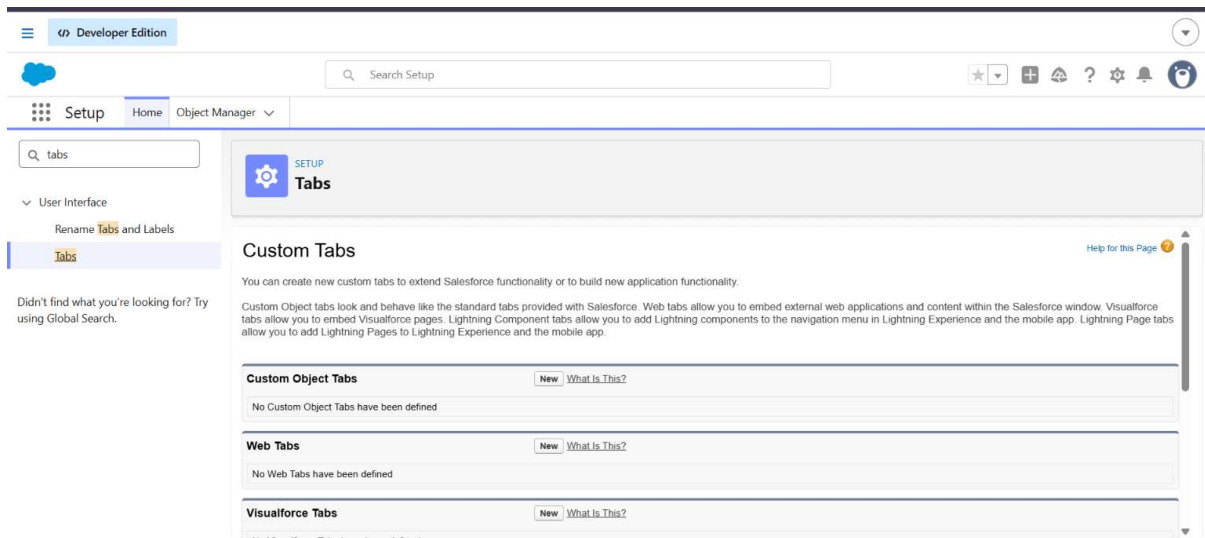
- Allow Sharing:** ☒
- Allow Bulk API Access:** ☒
- Allow Streaming API Access:** ☒

The 'Deployment Status' section contains the following fields:

- In Development:** ☐
- Deployed:** ☒

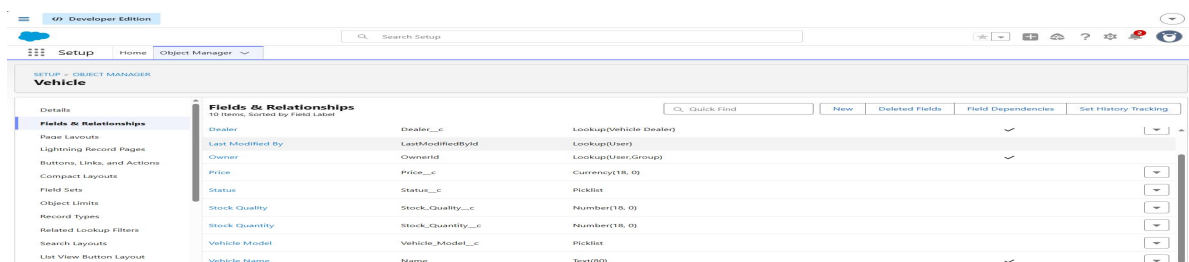
The 'Search Status' section contains the following fields:

- When this setting is enabled, your users can find records of this object type when they search.** [Learn more.](#)



2.2.2 Fields

Custom fields were created according to the information required by each object.



Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Vehicle Dealer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Dealer Code	Dealer_Code__c	Auto Number		
Dealer Email	Dealer_Email__c	Email		
Dealer Location	Dealer_Location__c	Text(18)		
Dealer Name	Name	Text(80)		✓
Dealer Phone	Dealer_Phone__c	Phone		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Vehicle Service Request

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

9 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Issue Description	Issue_Description__c	Text(18)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Service Date	Service_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Service Request Name	Name	Text(80)		✓
Vehicle Service Request Status	Vehicle_Service_Request_Status__c	Picklist		

Developer Edition

Search Setup

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Vehicle Test Drive

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find

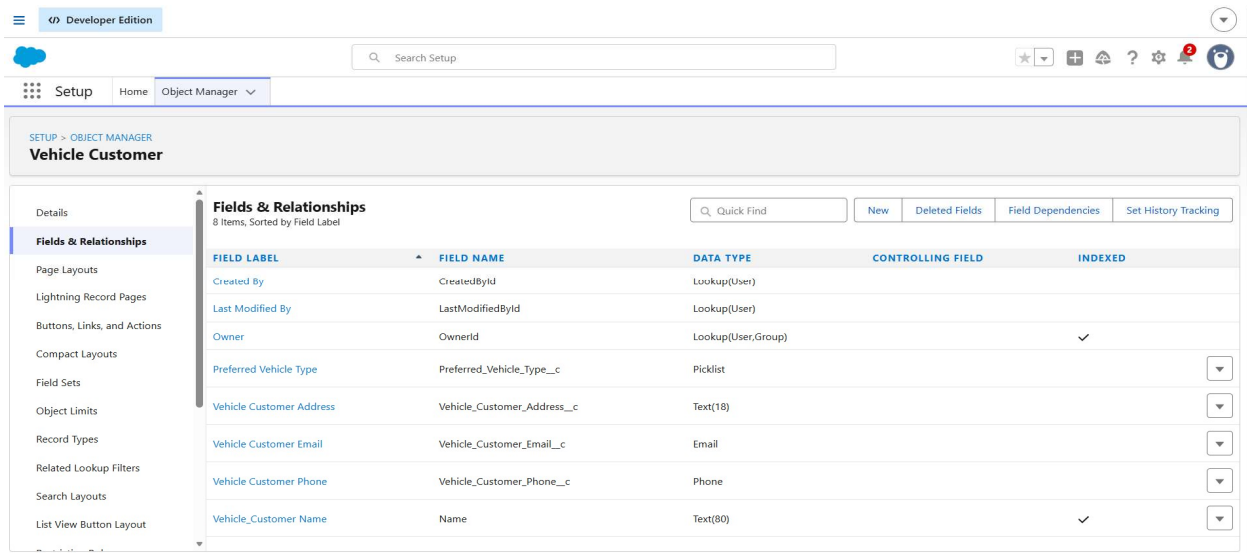
New

Deleted Fields

Field Dependencies

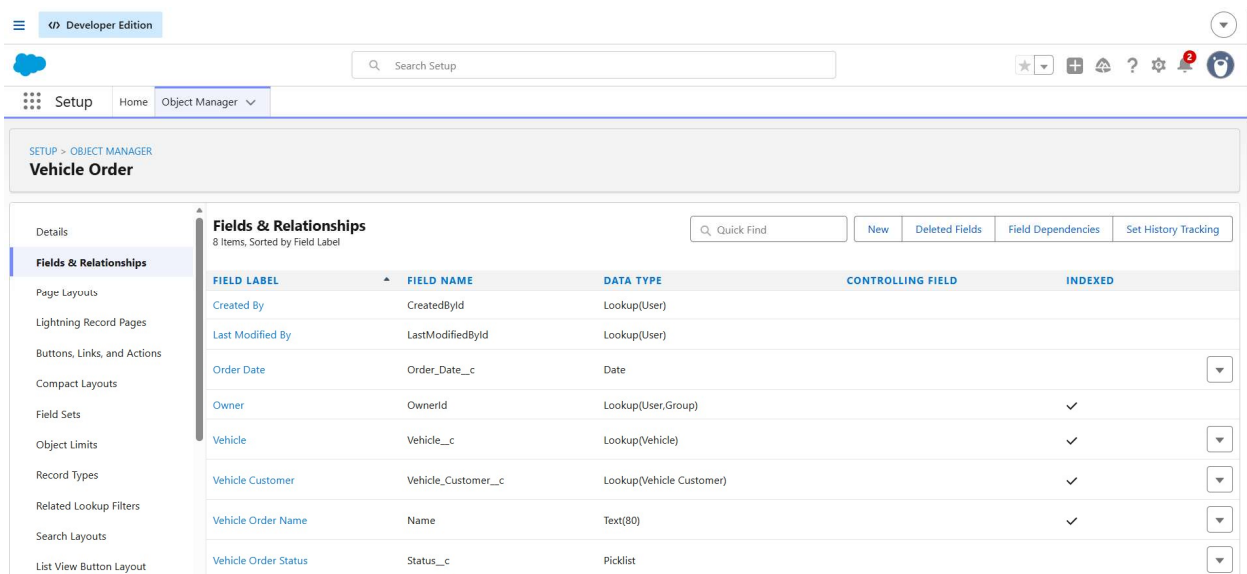
Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Test Drive Date	Test_Drive_Date__c	Date		
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Test Drive Name	Name	Text(80)		✓
Vehicle Test Drive Status	Vehicle_Test_Drive_Status__c	Picklist		



The screenshot shows the Salesforce Setup interface in Developer Edition, specifically the Object Manager for the 'Vehicle Customer' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The 'Fields & Relationships' section is selected, displaying a table of 8 items sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are Created By, Last Modified By, Owner, Preferred Vehicle Type, Vehicle Customer Address, Vehicle Customer Email, Vehicle Customer Phone, and Vehicle Customer Name.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Preferred Vehicle Type	Preferred_Vehicle_Type__c	Picklist		
Vehicle Customer Address	Vehicle_Customer_Address__c	Text(18)		
Vehicle Customer Email	Vehicle_Customer_Email__c	Email		
Vehicle Customer Phone	Vehicle_Customer_Phone__c	Phone		
Vehicle Customer Name	Name	Text(80)		✓



The screenshot shows the Salesforce Setup interface in Developer Edition, specifically the Object Manager for the 'Vehicle Order' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The 'Fields & Relationships' section is selected, displaying a table of 8 items sorted by Field Label. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are Created By, Last Modified By, Order Date, Owner, Vehicle, Vehicle Customer, Vehicle Order Name, and Vehicle Order Status.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Order Date	Order_Date__c	Date		
Owner	OwnerId	Lookup(User,Group)		✓
Vehicle	Vehicle__c	Lookup(Vehicle)		✓
Vehicle Customer	Vehicle_Customer__c	Lookup(Vehicle Customer)		✓
Vehicle Order Name	Name	Text(80)		✓
Vehicle Order Status	Status__c	Picklist		

2.2.3 Validation Rule-Vehicle Stock

A validation mechanism was implemented to ensure that an order cannot be confirmed when the associated vehicle does not have the required stock.

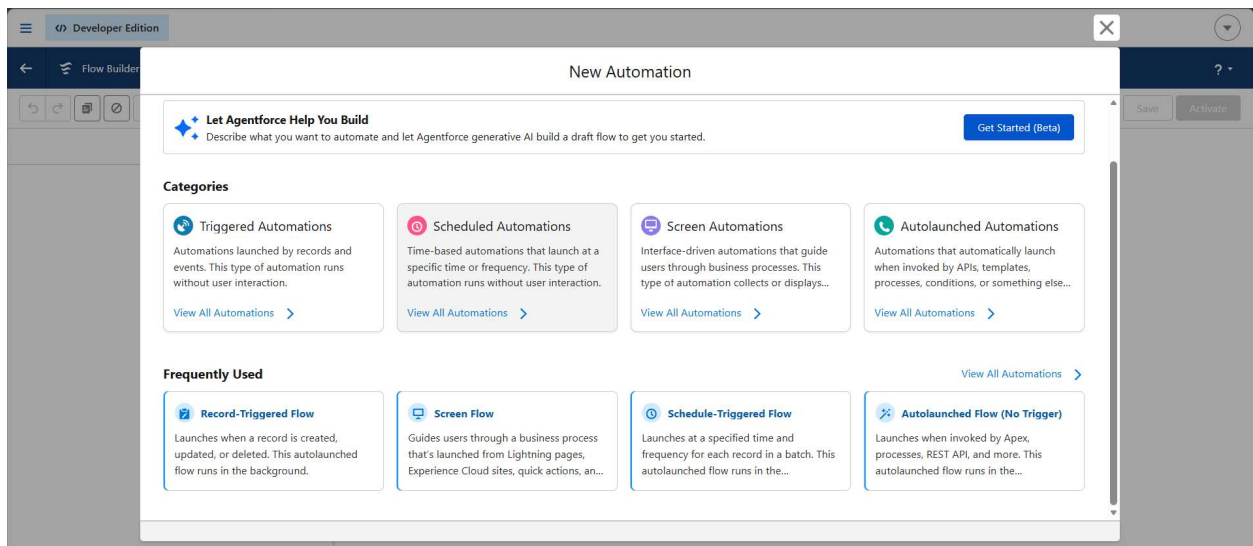
This rule protects the ordering process from an invalid stock condition.

For example, when the stock quantity of a selected vehicle is zero, an attempt to create a confirmed order should result in the configured validation response.

2.2.4 Flow Automation

Salesforce Flow was used to handle selected business processes without requiring every operation to be performed manually.

The project includes automation associated with dealer assignment and test-drive reminders.

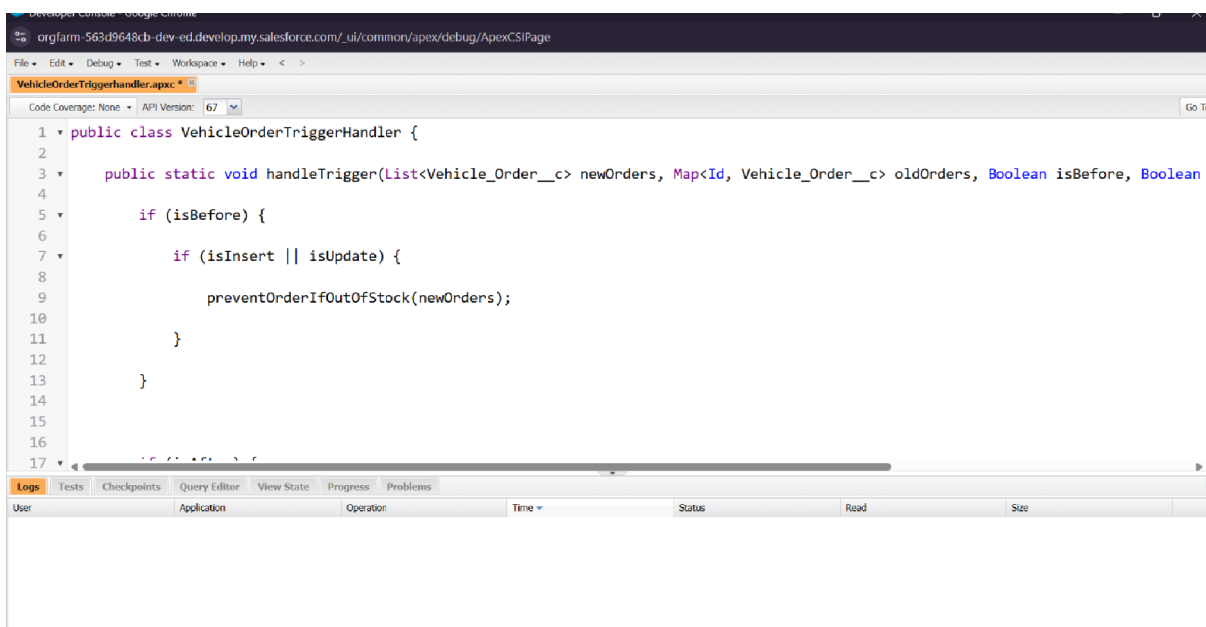
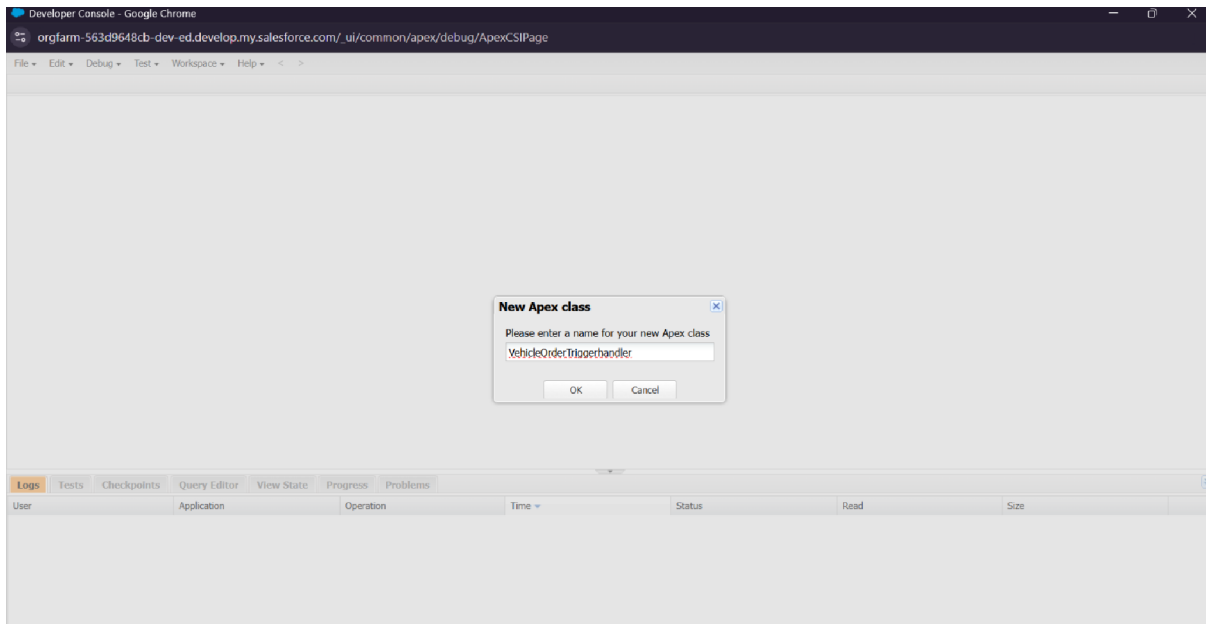


2.3 Apex Classes and Triggers

2.3.1 Apex Classes

- **Vehicle Order Processing:** Apex classes are used to handle the business logic associated with vehicle orders.
- **Stock Validation:** The classes check vehicle availability and ensure that orders follow the defined stock conditions.
- **Order Management:** Apex logic supports processing and updating vehicle order records based on their status.
- **Reusable Business Logic:** The business logic is separated into Apex classes to make the implementation easier to maintain and test.

- **Automation Support:** Apex classes work with triggers and other Salesforce automation to perform required operations efficiently.



2.3.2 Apex Trigger

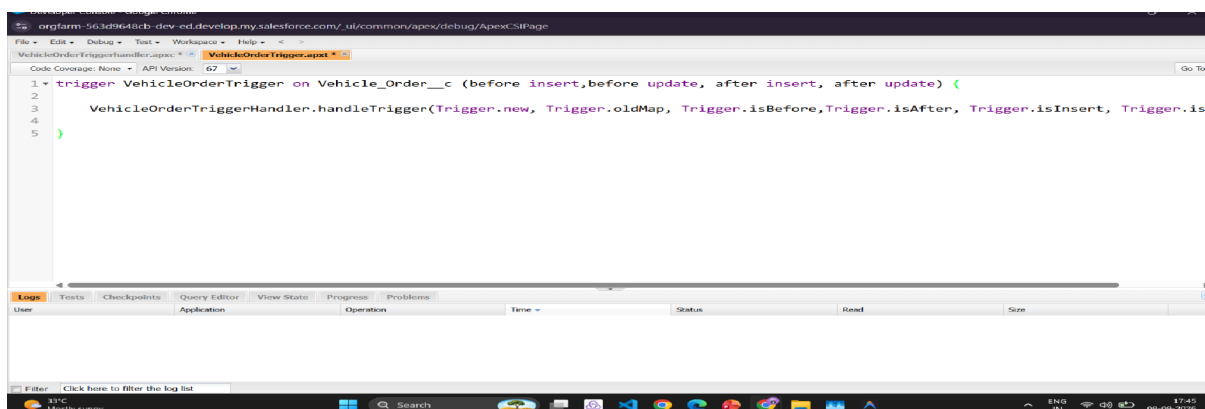
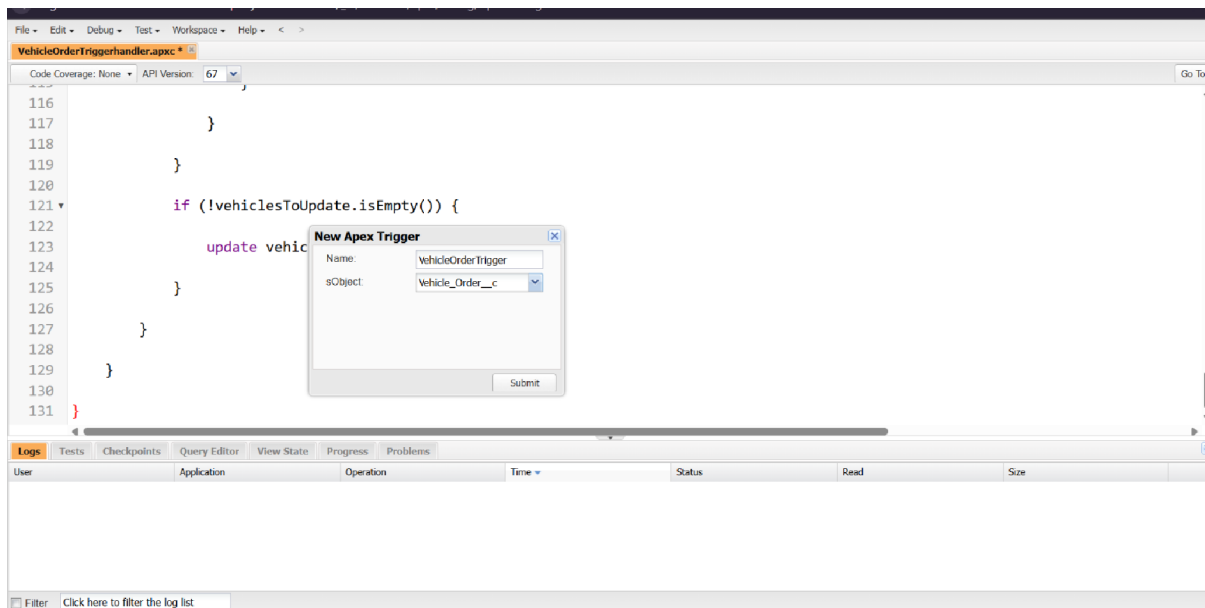
The vehicle order process also includes Apex-based backend logic.

The trigger responds to relevant order events and performs the configured business checks.

The main responsibilities include:

- Checking stock availability.
- Preventing invalid order processing.
- Updating vehicle stock after a confirmed transaction.
- Supporting order-status processing.

The trigger logic is organized using a handler approach so that business logic can be separated from the trigger definition.

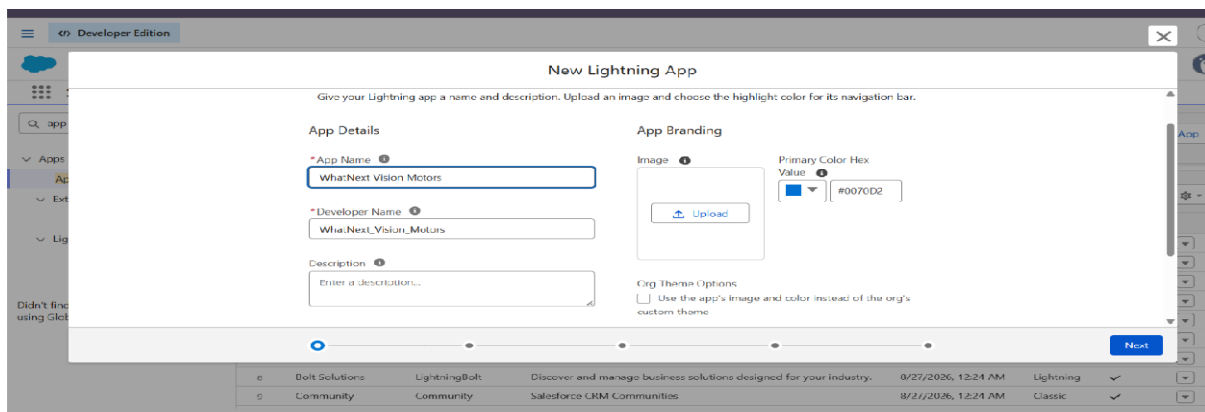
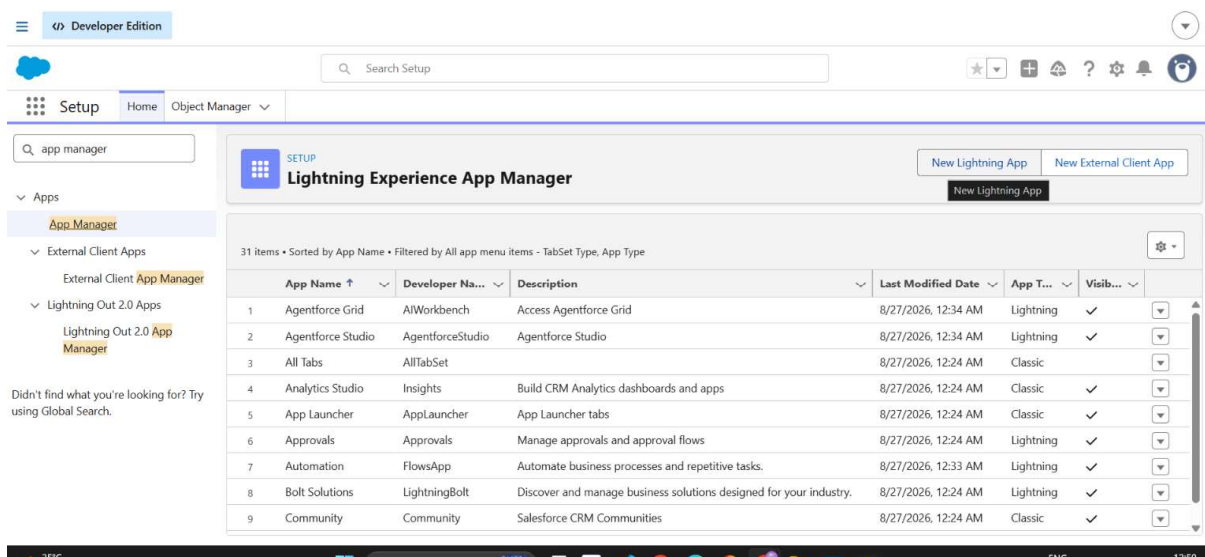


Phase 3: UI/UX Development & Customization

The third phase concentrated on creating a practical user interface within Salesforce Lightning. The project guidelines specifically require the Lightning App, App Manager, page layouts, Dynamic Forms, user management, reports, dashboards, Lightning Pages, and LWC where applicable.

3.1 Lightning App setup through App Manager

The Lightning App provides a centralized interface for managing Vehicles, Dealers, Customers, Orders, Test Drives, and Service Requests. It improves navigation and provides users with easy access to the required Salesforce records and business functions.



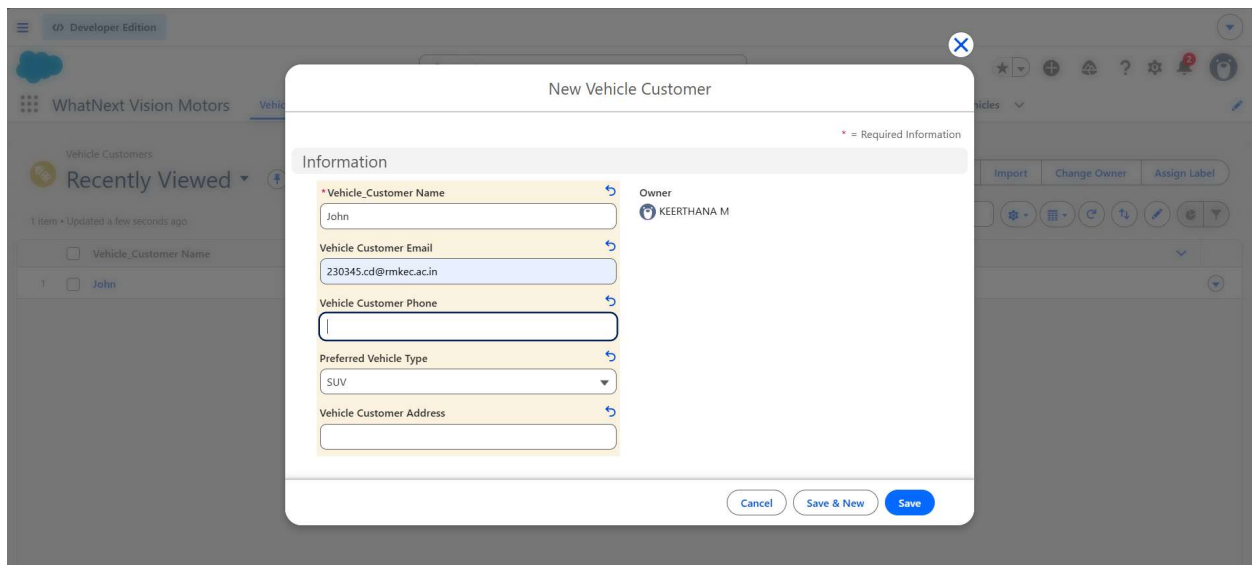
3.2 Page Layouts & Dynamic Forms

Page layouts were configured to arrange the fields and related information displayed on records. The layout design gives users access to relevant information while keeping the record interface organized.

Dynamic Forms were used where conditional field visibility was required. Instead of presenting every field at the same time, fields can be displayed according to the values or conditions associated with a record.

3.3 User Management

Users were configured according to the access requirements of the application.

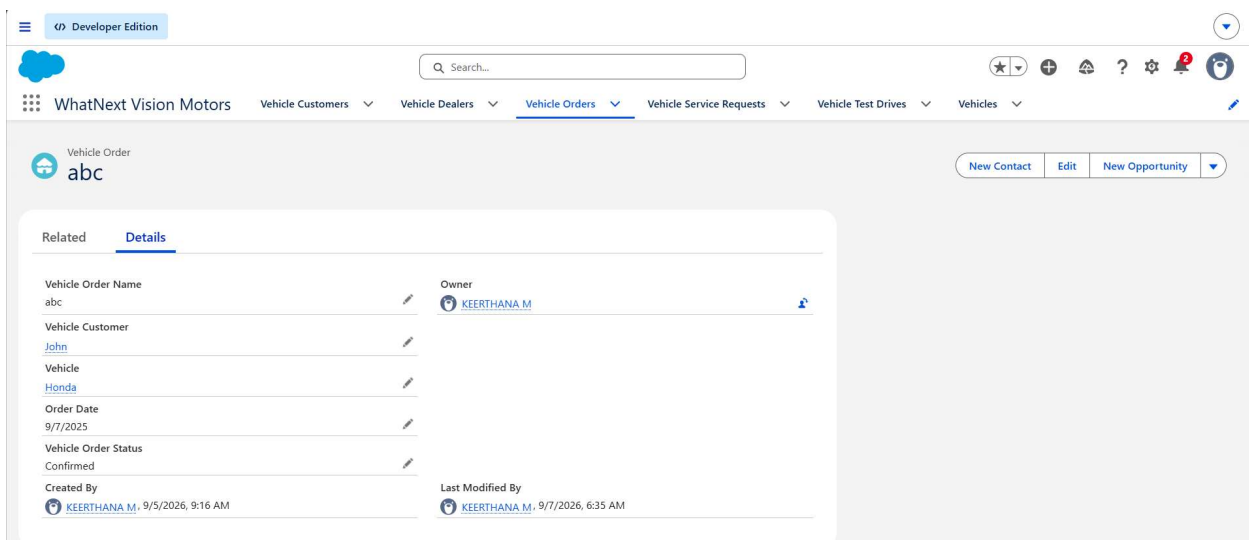
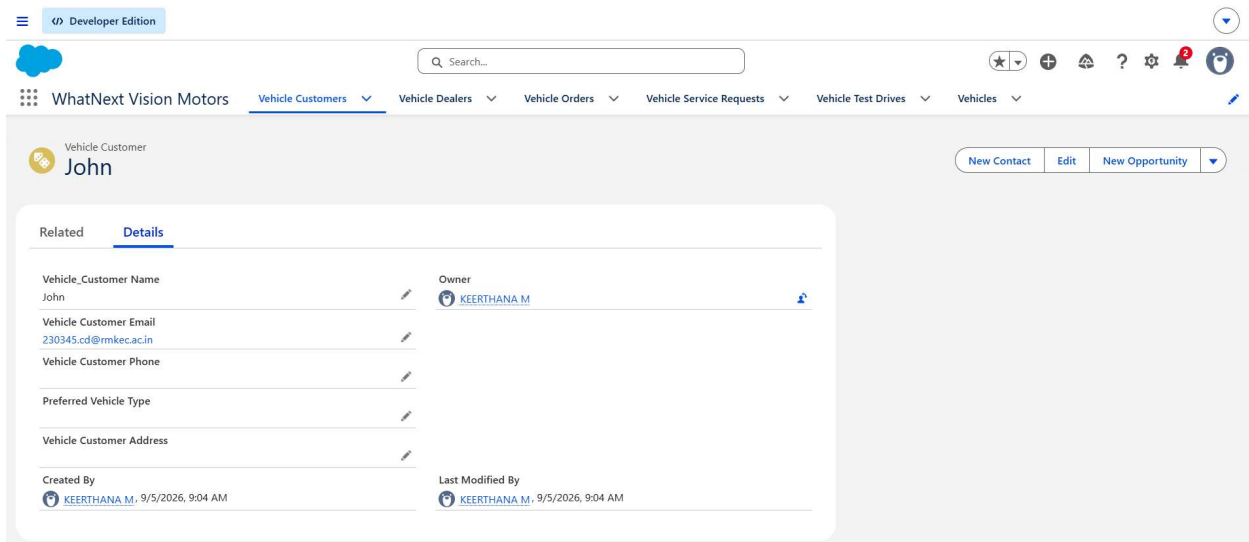


The screenshot shows a 'New Vehicle Customer' form in a Salesforce application. The form is titled 'New Vehicle Customer' and includes a close button (X) in the top right corner. A legend indicates that an asterisk (*) denotes required information. The form is divided into two main sections: 'Information' and 'Owner'. The 'Information' section contains several fields: 'Vehicle_Customer Name' (required, with the value 'John'), 'Vehicle Customer Email' (with the value '230345.cd@rmkec.ac.in'), 'Vehicle Customer Phone' (empty), 'Preferred Vehicle Type' (a dropdown menu with 'SUV' selected), and 'Vehicle Customer Address' (empty). The 'Owner' section shows a user profile for 'KEERTHANA M'. At the bottom of the form, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The background of the application shows a sidebar with 'WhatNext Vision Motors' and a 'Recently Viewed' list containing one item, 'John'.

3.4 Reports and Dashboards

Reports were configured to provide structured views of Salesforce records.

Dashboards provide a visual representation of selected report information. They help users obtain a quick understanding of business activity without manually reviewing individual records.

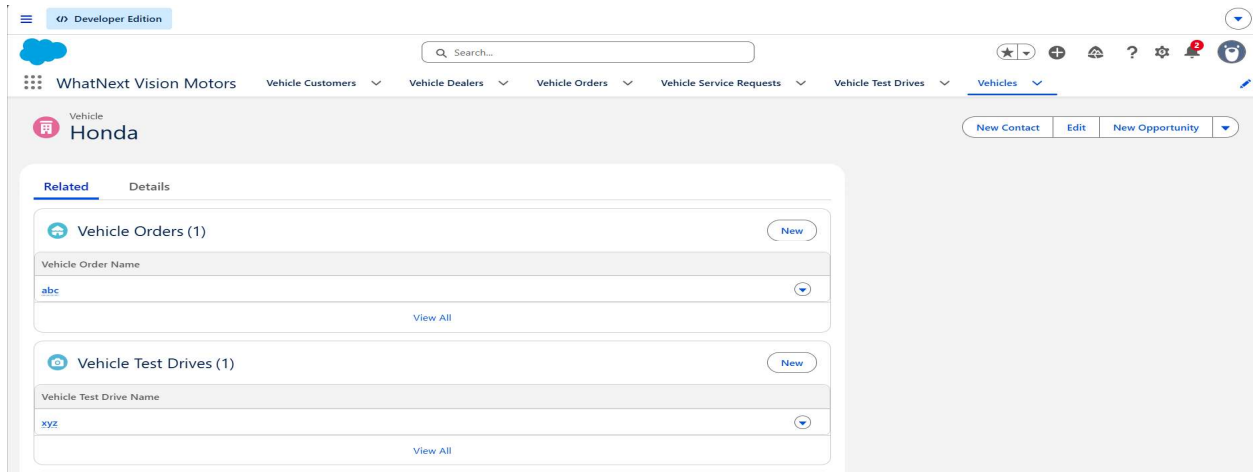


3.5 Lightning Record Pages

Lightning Record Pages were configured to provide a useful view of individual records.

Relevant components may include:

- Record details
- Highlights panel
- Related lists



Phase 4: Data Migration, Testing & Security

The fourth phase focused on populating the CRM with data, verifying the implemented functionality, and configuring security controls. The guidelines specifically require documentation of data loading, history tracking, duplicate/matching rules, profiles, roles, permission sets, sharing rules, and testing.

4.1 Data Loading

Initial data was prepared in a structured format before being added to Salesforce.

The data-loading process involved:

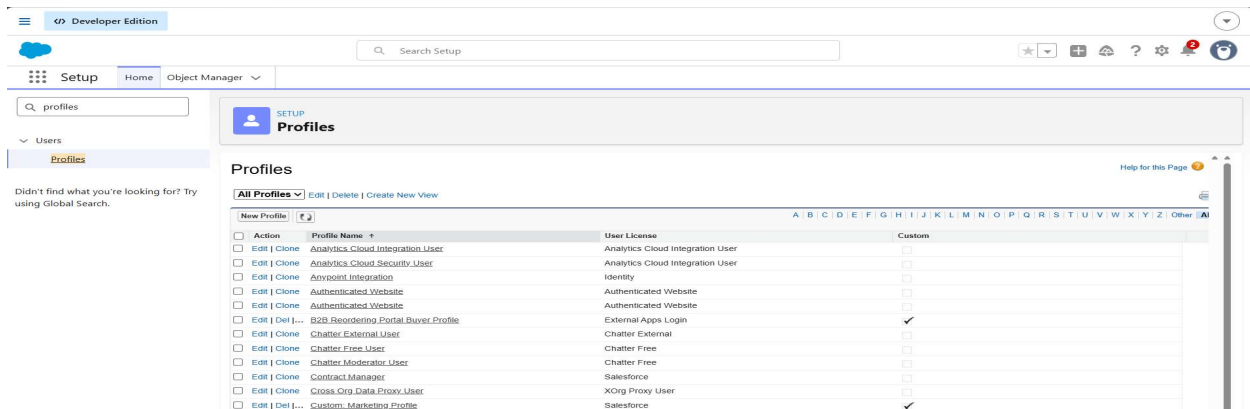
1. Preparing sample records.
2. Mapping the data fields to Salesforce fields.
3. Loading the required records.
4. Checking whether the records were created correctly.
5. Verifying relationships between associated records.

Where applicable, Salesforce Data Import Wizard or Data Loader can be used depending on the object and quantity of records.

4.2 Profiles

Profiles provide the basic permission structure for Salesforce users.

They determine the level of access users have to applications, objects, fields, and other Salesforce functionality.



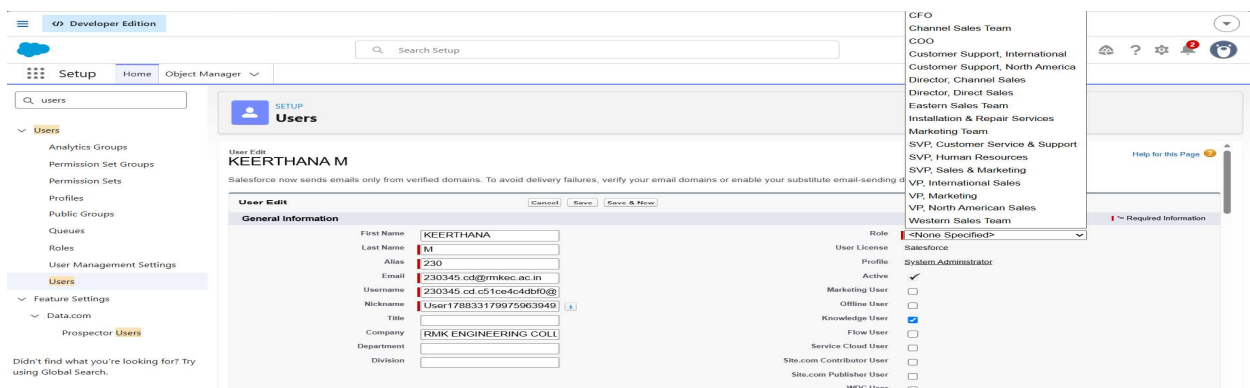
4.3 Roles and Role Hierarchy

The role hierarchy represents the organizational structure used for record-access considerations.

A possible structure for the project is:

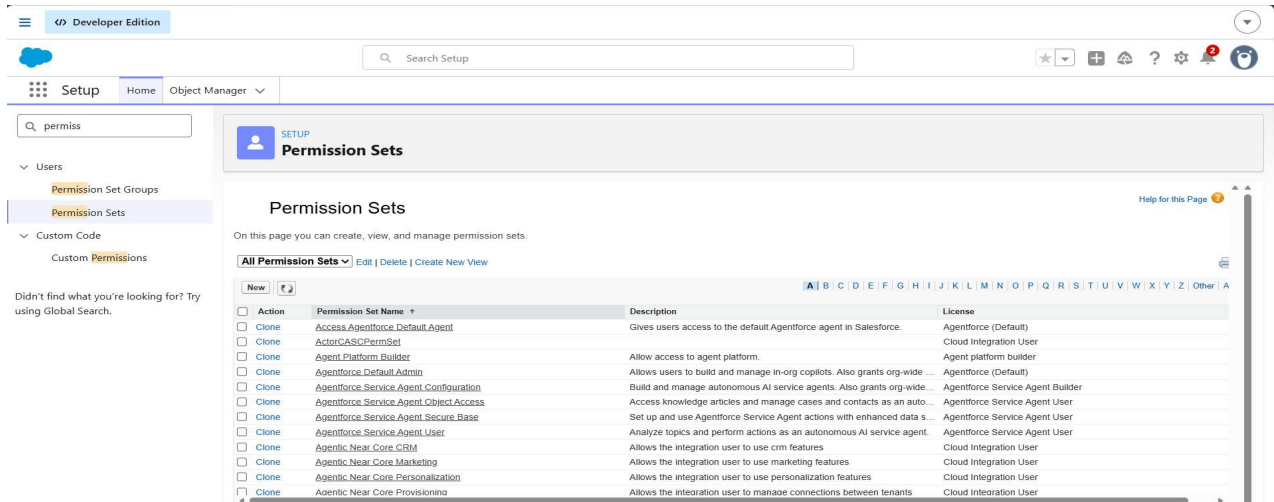
CEO
↓
Sales Manager
↓
Sales Representative

The actual hierarchy shown in the Salesforce org should be used in the final screenshot.



4.4 Permission Sets

Permission Sets were used to provide additional access to users who require specific Salesforce functionality.



4.5 Sharing Rules

Sharing configuration determines how records can be accessed by different users.

The sharing model should be aligned with the organization's responsibilities so that users can work with the information required for their role while avoiding unnecessary exposure of restricted records.

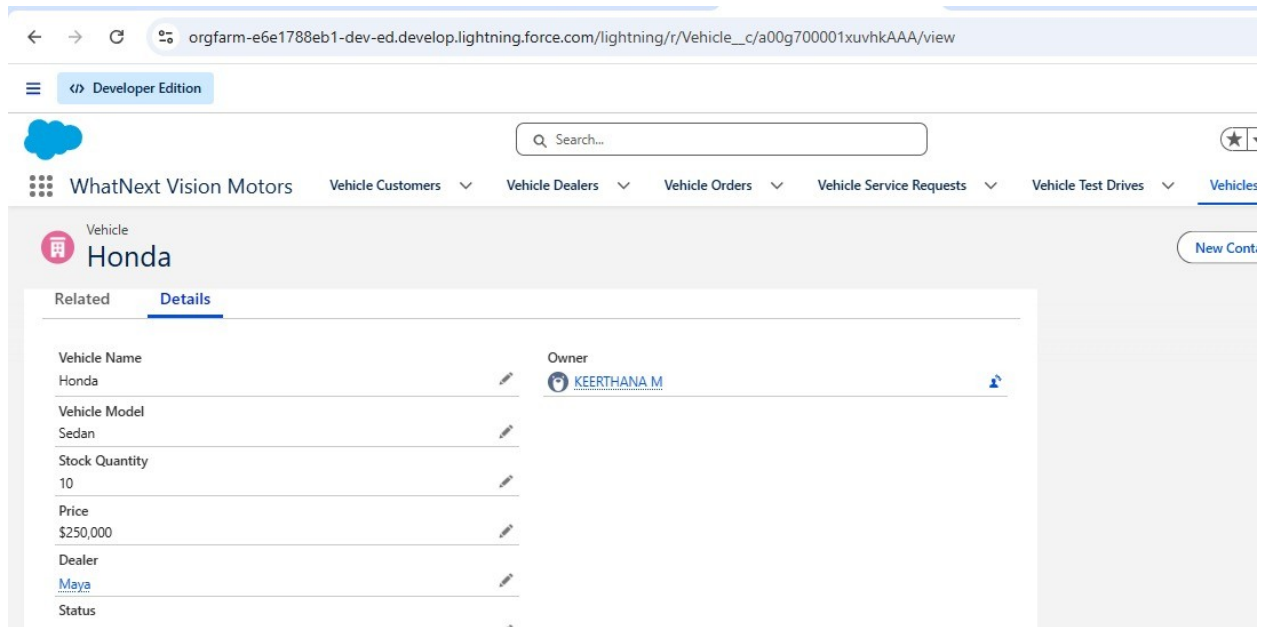
4.6 Preparation of test cases for each and every salesforce features like booking creation, Approval Process, Automatic Task creation, flows, triggers etc must be submitted along with input and output Screenshots.

✓ Test Case 1 – Vehicle Creation

Verify that a valid vehicle record can be created.

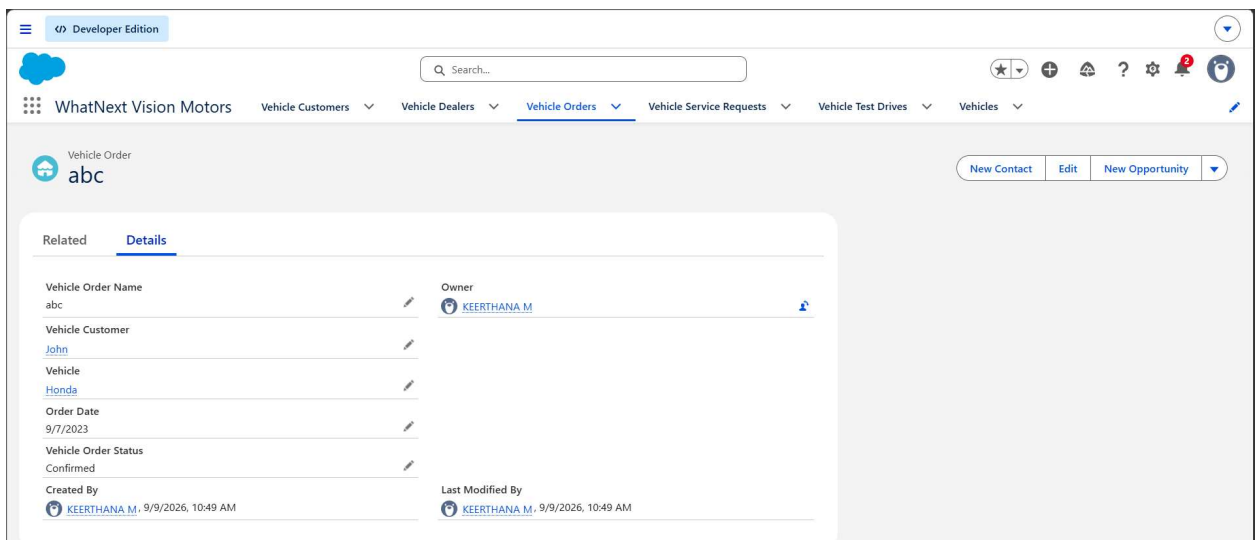
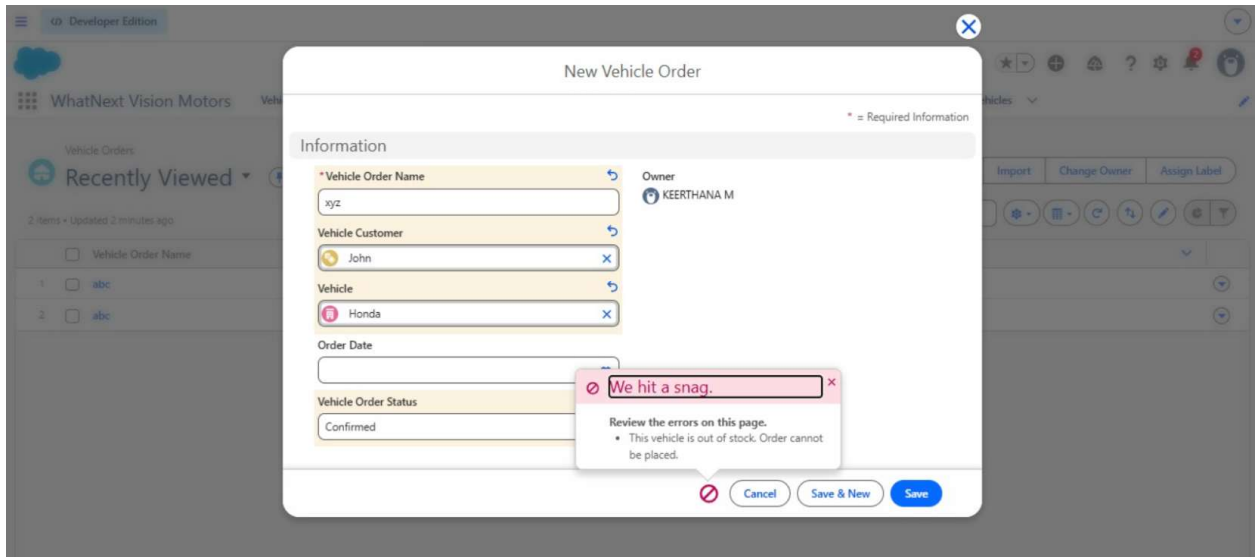
- Vehicle Name: Honda
- Model: Sedan
- Stock Quantity: 1

- Price: ₹2,50,000
- Status: Available
- Dealer: Maya



✓ Test Case 2 – Out-of-Stock Order

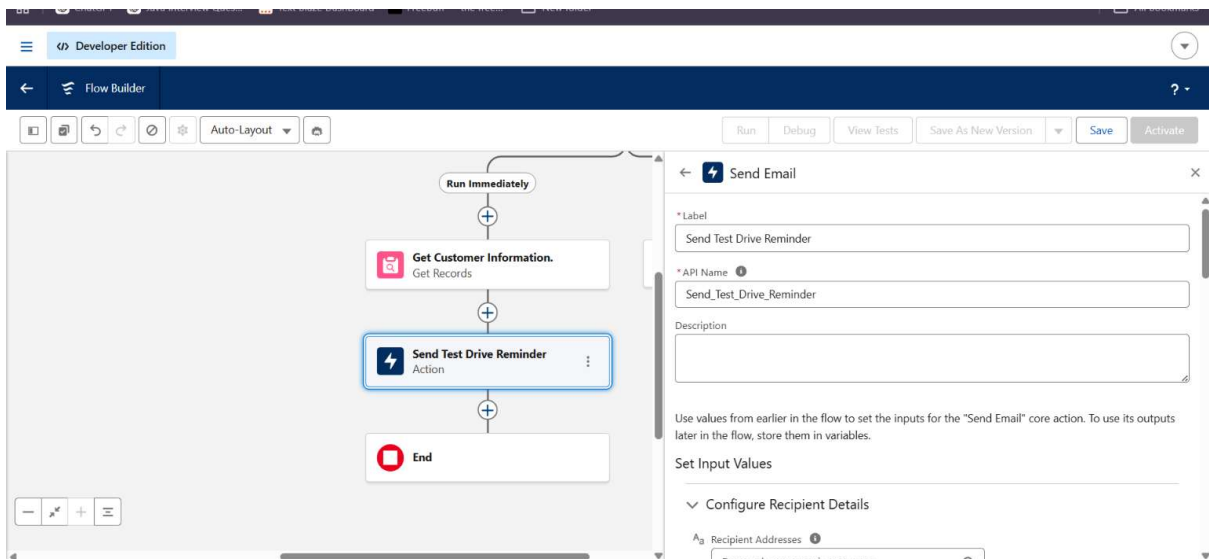
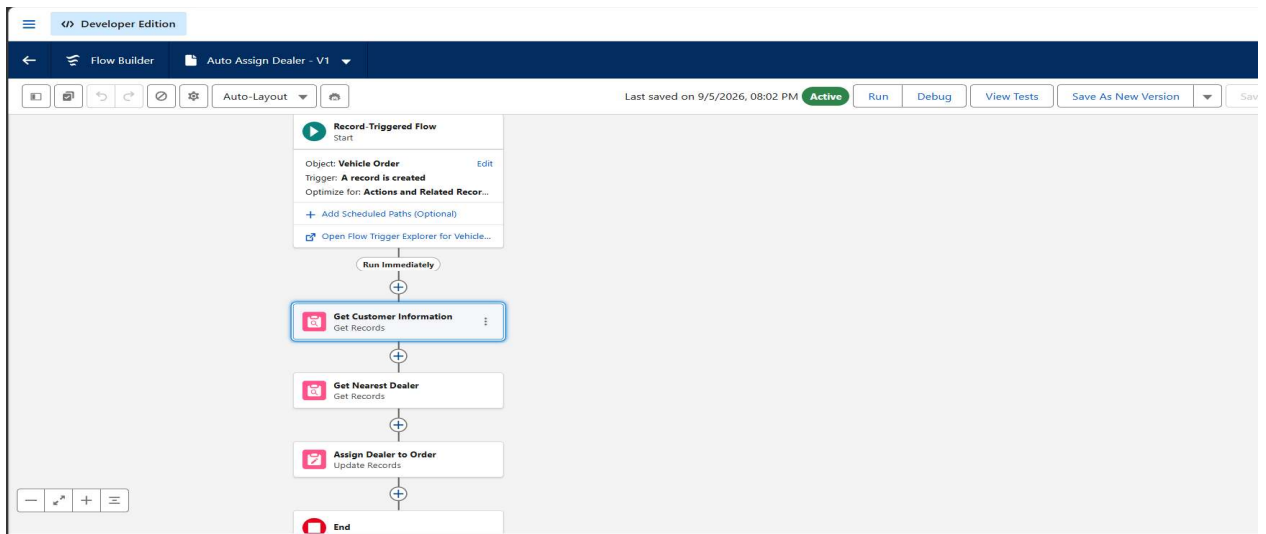
Verify that an order cannot be confirmed when the selected vehicle has no available stock.

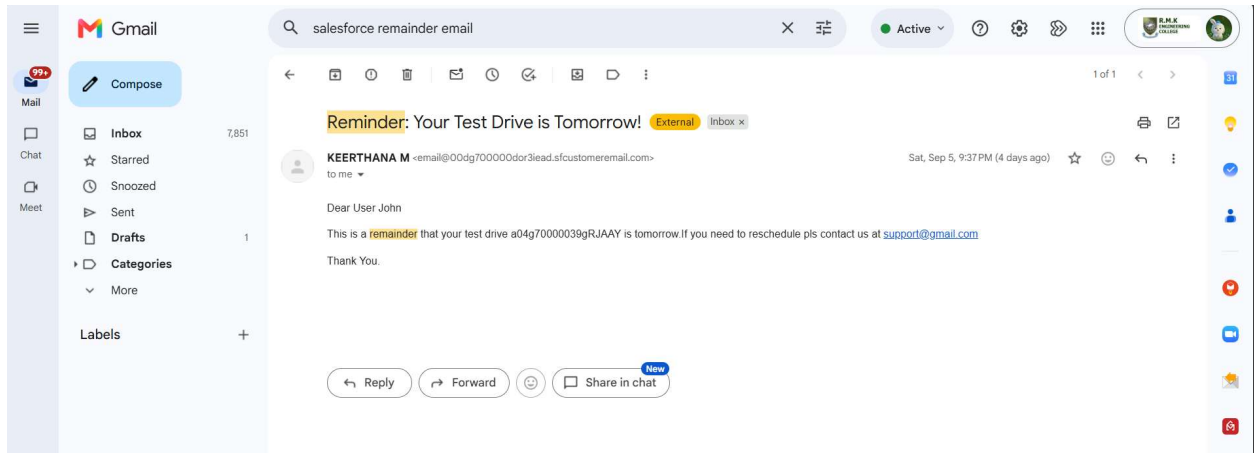


✓ Test Case 3 – Test Drive Remainder Email

Verify the automated reminder process for an upcoming test drive.

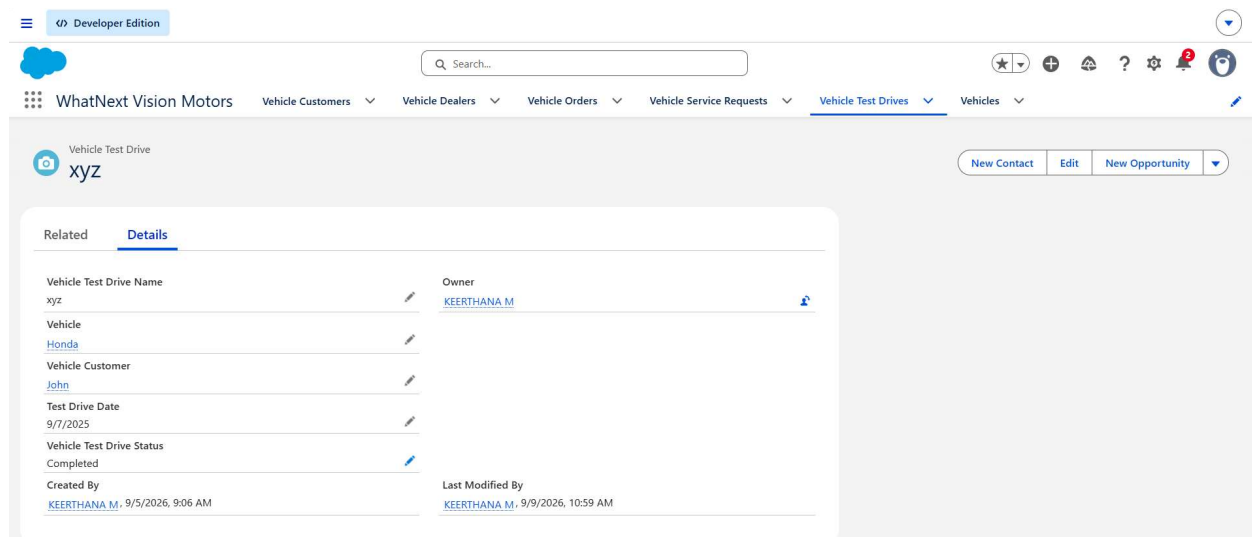
- Customer: Existing customer
- Customer Email: Valid email
- Status: Scheduled
- Test Drive Date: Following day





✓ Test Case 4 – Pending Order Batch Processing

Verify that a pending order can be processed after stock becomes available.



Phase 5: Deployment, Documentation & Maintenance

The final development phase deals with moving the Salesforce components to the intended environment, monitoring the application, handling issues, and maintaining the system. The guidelines specifically require a deployment strategy, maintenance approach, monitoring, and troubleshooting process.

5.1 Deployment Strategy

Salesforce metadata can be transferred between environments using supported deployment mechanisms.

For this project, the deployment process can include the following components:

- Custom objects
- Custom fields
- Relationships
- Validation rules
- Flows
- Apex classes
- Apex triggers
- Lightning pages
- Reports

5.2 Basic description of how the system will be maintained and monitored

Salesforce monitoring tools can help administrators identify problems in automated processes.

Important areas include:

Apex Jobs

Used to monitor batch and scheduled Apex execution.

Debug Logs

Useful for investigating unexpected behaviour in Apex and other Salesforce operations.

Flow Errors

Flow failure information can help identify problems in automated processes.

User Access

Periodic permission reviews help ensure that users retain only the access they need.

5.3 Documentation of troubleshooting approach

A structured troubleshooting method can be followed when an issue is identified.

Step 1 – Identify the Problem

Determine which Salesforce feature is producing the unexpected result.

Step 2 – Reproduce the Issue

Repeat the same operation using controlled test data.

Step 3 – Review Configuration

Inspect the relevant object, fields, validation rules, Flow, permissions, or Apex logic.

Step 4 – Check Logs

When programmatic or automated processing is involved, review available debugging information.

Step 5 – Correct the Issue

Modify the configuration or code responsible for the problem.

Step 6 – Retest

Run the original scenario again and verify that the expected result is obtained.

Step 7 – Deploy the Correction

After successful testing, move the corrected configuration to the required environment using the chosen deployment p

Conclusion

The WhatNext Vision Motors project resulted in a Salesforce CRM environment capable of organizing several automotive business activities within a common platform.

The implementation demonstrates how Salesforce can be customized to represent business-specific entities and processes.

The major outcomes of the project are:

- Automotive records are organized using Salesforce objects.
- Relationships connect customers, vehicles, dealers, orders, and related activities.
- Vehicle stock conditions can be checked during order processing.
- Dealer assignment can be automated.
- Test-drive reminders can be generated through automation.
- Pending orders can be processed using background Apex functionality where implemented.
- Lightning pages provide a structured user experience.
- Security controls provide controlled access.
- Reports and dashboards support business monitoring.
- Testing provides evidence that the implemented functionality works according to the defined scenarios.

The project also provides practical experience in combining Salesforce declarative tools with Apex development.